

SMART LOST & FOUND MANAGEMENT SYSTEM

1. INTRODUCTION

The Smart Lost & Found Management System is a ServiceNow-based application developed to simplify and automate the management of lost and found items.

In many organizations and educational institutions, lost-and-found activities are handled manually through registers, spreadsheets, messages, or informal communication. Such approaches make it difficult to maintain accurate records, identify matching items, verify claims, track returns, and communicate updates to users.

The proposed system provides a centralized platform where users can report lost items and found items. Administrators can manage these records, identify potential matches, verify claims, process returns, and communicate important updates through automated notifications.

The application uses ServiceNow App Engine Studio for application development and Flow Designer for workflow automation. A custom Service Portal experience provides users with dedicated forms for reporting lost and found items.

The system manages the complete lifecycle of an item from reporting to matching, claiming, verification, approval, return, and notification.

2. PROBLEM STATEMENT

The traditional lost-and-found process is often manual and lacks a centralized system for managing information.

The major problems associated with manual lost-and-found management include:

- Difficulty maintaining organized records of lost and found items.
- Difficulty identifying whether a newly found item matches an existing lost item.
- Lack of a structured claim verification process.
- Manual tracking of item returns.
- Delayed communication with users.
- Difficulty monitoring the complete lifecycle of a lost item.
- Increased administrative effort.

Therefore, there is a need for a centralized and automated system that can manage lost and found items efficiently while providing proper tracking, verification, and notification mechanisms.

3. OBJECTIVES

The main objectives of the Smart Lost & Found Management System are:

1. To provide a centralized platform for managing lost and found items.
2. To allow users to report lost items through a structured form.

3. To allow users to report found items through a structured form.
4. To maintain organized records of lost items, found items, claims, and returns.
5. To identify potential matches between lost and found items.
6. To automate the claim verification process.
7. To automatically create return records when claims are approved.
8. To automate notifications for important claim and return events.
9. To implement role-based access control.
10. To provide a user-friendly Service Portal interface.
11. To reduce repetitive manual work through workflow automation.

4. PROPOSED SOLUTION

The proposed solution is a ServiceNow application that centralizes the entire lost-and-found management process.

The system consists of four major data modules:

- Lost Items
- Found Items
- Claims
- Item Returns

When a user reports an item as lost, the information is stored in the Lost Items table. When an item is found, it is recorded in the Found Items table.

The system then uses Flow Designer to identify potential matches using attributes such as category, color, and brand.

When a user submits a claim, the claim is automatically moved into the verification process. Once the claim is approved, the system automatically creates an Item Return record. After the item is returned, the user receives an automated email notification.

Therefore, the application provides an end-to-end workflow:

Report → Match → Claim → Verify → Approve → Return → Notify

5. SCOPE OF THE PROJECT

The system is designed to support lost-and-found operations within an organization, educational institution, office, or similar environment.

The current scope includes:

- Reporting lost items.
- Reporting found items.
- Maintaining item information.

- Managing potential item matches.
- Submitting claims.
- Verifying claims.
- Approving or rejecting claims.
- Creating return records.
- Managing return status.
- Sending automated email notifications.
- Providing role-based access.
- Providing a custom Service Portal interface.
- Automating business processes using ServiceNow Flow Designer.

The application has been developed and tested in a ServiceNow Personal Developer Instance.

6. TECHNOLOGY STACK

The following ServiceNow technologies were used to develop the application:

ServiceNow:

Used as the primary application development and workflow platform.

App Engine Studio:

Used to create and configure the custom application, tables, roles, and application components.

Flow Designer:

Used to automate claim processing, item matching, return processing, and notifications.

Service Portal / Experience:

Used to provide the user-facing application interface.

UI Builder:

Used to create and configure custom portal pages and reporting forms.

ServiceNow Tables:

Used to store lost items, found items, claims, and return information.

Reference Fields:

Used to establish relationships between application records.

Roles and Permissions:

Used to control access to application functionality.

Email Notifications:

Used to automatically communicate important status changes to users.

ServiceNow PDI:

Used as the development and testing environment.

7. SYSTEM ARCHITECTURE

The Smart Lost & Found Management System follows a centralized workflow-based architecture.

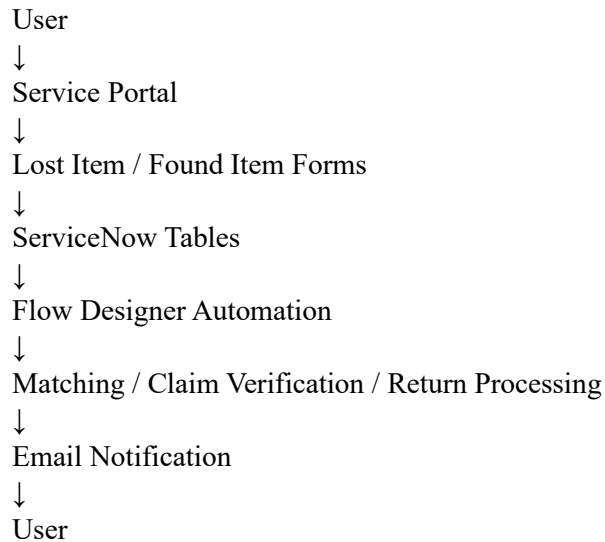
The user interacts with the system through the Service Portal. The submitted information is stored in the corresponding ServiceNow tables.

The backend consists of four primary tables:

Lost Items, Found Items, Claims, and Item Returns.

Flow Designer connects these modules and automates the business logic.

The overall architecture can be represented as follows:



The architecture enables information to move systematically from one stage of the process to another.

8. APPLICATION MODULES

8.1 Lost Items Module

The Lost Items module stores information about items reported as lost.

The table contains the following fields:

- Item Name
- Category
- Description
- Color
- Brand
- Lost Date
- Lost Location
- Reported By
- Status
- Priority
- Contact Information

The Lost Items table uses an auto-numbering system with the prefix "LST".

Example:

LST0001010

The available statuses are:

- Reported
- Searching
- Potential Match
- Claimed
- Returned
- Closed

Priority levels include:

- 1 - Critical
- 2 - High
- 3 - Moderate
- 4 - Low

8.2 Found Items Module

The Found Items module stores information about items that have been found.

The table contains:

- Item Name
- Category
- Description
- Color
- Brand
- Found Date
- Found Location
- Found By
- Storage Location
- Status

The Found Items table uses the prefix "FND".

Example:

FND0001009

The available statuses are:

- Found
- Under Verification
- Potential Match
- Claimed
- Returned
- Closed

8.3 Claims Module

The Claims module manages requests made by users to claim found items.

The Claims table contains:

- Lost Item
- Found Item
- Claimant
- Claim Description
- Proof Description
- Claim Status
- Verification Notes
- Claim Date

Each claim receives an auto-generated identifier beginning with "CLM".

Example:

CLM0001004

The claim status values are:

- Submitted
- Under Verification
- Approved
- Rejected
- Completed

8.4 Item Returns Module

The Item Returns module manages the final return of an item to the verified claimant.

The table contains:

- Claim
- Lost Item
- Found Item
- Returned To
- Return Date
- Return Status
- Handover Notes

Each return receives an identifier beginning with "RET".

Example:

RET0001007

The return status values are:

- Pending
- Ready for Pickup
- Returned
- Cancelled

9. DATABASE AND TABLE DESIGN

The application consists of four interconnected custom tables.

Lost Items Table

Technical Name:

x_2235078_smart_0_lost_items

This table stores information about items reported as lost.

Found Items Table

Technical Name:

x_2235078_smart_0_found_item_records

This table stores information about items reported as found.

Claims Table

Technical Name:

x_2235078_smart_0_claim_records

This table stores claim information and connects the claimant with the corresponding lost and found items.

Item Returns Table

Technical Name:

x_2235078_smart_0_item_return_records

This table stores information about the final return of the item.

The relationships between the tables allow the complete lifecycle of an item to be tracked.

For example:

Lost Item → Found Item → Claim → Return

10. ROLE-BASED ACCESS CONTROL

Two application roles were created for the system.

10.1 Administrator Role

Role:

x_2235078_smart_0.admin

The administrator manages the application and performs activities such as:

- Managing lost and found records.
- Verifying claims.
- Approving or rejecting claims.
- Managing return records.
- Monitoring application workflows.
- Managing application data.

10.2 User Role

Role:

x_2235078_smart_0.user

The application user can perform activities such as:

- Reporting lost items.
- Reporting found items.
- Submitting claims.
- Interacting with the lost-and-found process.

Role-based access ensures that application functionality is provided according to the responsibilities of each user type.

11. WORKFLOW AUTOMATION

ServiceNow Flow Designer is a major component of the application.

Six flows were implemented to automate the major business processes.

11.1 Process New Lost Item Claim

Purpose

This flow automatically processes a newly created claim.

Trigger

Claims record is created.

Action

The Claim Status is automatically changed to:

Under Verification

Workflow

New Claim Created

↓

Flow Triggered

↓

Claim Status Updated

↓

Under Verification

This ensures that every newly submitted claim enters the verification process automatically.

12. FIND POTENTIAL LOST ITEM MATCH

Purpose

This flow identifies potential matches between newly reported found items and existing lost items.

Trigger

Found Item record is created.

Matching Criteria

The system looks up Lost Items based on:

- Category
- Color
- Brand

The values are dynamically taken from the newly created Found Item record.

Workflow

Found Item Created

↓

Read Category

↓

Read Color

↓

Read Brand

↓

Look Up Lost Items

↓

Process Matching Records

↓

Update Lost Item Status

↓

Potential Match

This provides an automated first-level matching mechanism.

13. PROCESS ITEM RETURN

Purpose

This flow manages newly created Item Return records.

Trigger

Item Return record is created.

Action

The Return Status is automatically updated to:

Ready for Pickup

Workflow

Item Return Created

↓

Flow Triggered

↓
Return Status Updated

↓
Ready for Pickup

This provides a clear status to indicate that the item is ready to be collected.

14. CREATE RETURN FOR APPROVED CLAIM

Purpose

This flow automatically creates an Item Return record after a claim has been approved.

Trigger

Claim record is updated.

Condition

Claim Status = Approved

Action

A new Item Return record is created.

The following information is mapped:

Claim → Trigger Claim

Lost Item → Trigger Lost Item

Found Item → Trigger Found Item

Returned To → Trigger Claimant

Return Status → Pending

Workflow

Claim Updated

↓
Check Claim Status

↓
Approved?

↓
Yes

↓
Create Item Return

↓
Return Status = Pending

This eliminates the need for the administrator to manually create a return record after approving a claim.

15. NOTIFY CLAIMANT WHEN CLAIM APPROVED

Purpose

This flow automatically notifies the claimant when their claim is approved.

Trigger

Claim record is updated.

Condition

Claim Status = Approved

Action

An email is sent to the claimant.

Email Subject

Your Lost & Found Claim Has Been Approved

Workflow

Claim Updated

↓

Claim Status = Approved

↓

Send Email

↓

Claimant Notified

This improves communication and reduces manual follow-up.

16. NOTIFY USER WHEN ITEM IS RETURNED**Purpose**

This flow automatically notifies the user after the item has been returned.

Trigger

Item Return record is updated.

Condition

Return Status = Returned

Action

An email is sent to the person listed in the Returned To field.

Email Subject

Your Lost & Found Item Has Been Returned

Workflow

Item Return Updated

↓

Return Status = Returned

↓

Send Email

↓

User Notified

The flow was successfully tested and a corresponding email record was generated in ServiceNow.

17. SERVICE PORTAL AND USER INTERFACE

A custom ServiceNow Portal experience was created for the application.

Portal Name:

Smart Lost & Found Management

The portal provides a user-facing interface for accessing the application functionality.

The portal includes customized navigation and reporting pages.

The major custom pages developed are:

- Report Lost Item
- Report Found Item

These pages allow users to submit information without directly interacting with the backend tables.

18. REPORT LOST ITEM PAGE

The Report Lost Item page was developed using UI Builder and ServiceNow Form Controller.

The Form Controller is configured to use the Lost Items table.

The user enters information such as:

- Item Name
- Category
- Description
- Color
- Brand
- Lost Date
- Lost Location
- Reported By
- Priority
- Contact Information

After completing the form, the user selects the Submit Lost Item action.

The record is then saved in the Lost Items table.

Workflow

User opens Report Lost Item page

↓

Enters item information

↓

Clicks Submit Lost Item

↓

Form Save Action

↓

Lost Item Record Created

The form was successfully tested by creating a Lost Item record.

19. REPORT FOUND ITEM PAGE

A separate Report Found Item page was created for recording found items.

The Form Controller is configured to use the Found Items table.

The user enters information about the found item and submits the form.

The information is then stored in the Found Items table.

Workflow

User opens Report Found Item page



Enters found item information



Submits form



Found Item Record Created

The form was successfully tested.

20. COMPLETE SYSTEM WORKFLOW

The complete workflow of the application is as follows:

User reports a lost item.



Lost Item record is created.



Another user reports a found item.



Found Item record is created.



Matching flow searches for corresponding Lost Items.



Potential match is identified.



User submits a claim.



Claim automatically moves to Under Verification.

↓

Administrator verifies the claim.

↓

If rejected, the claim is rejected.

↓

If approved, an Item Return record is created.

↓

Return is processed.

↓

Return Status becomes Ready for Pickup.

↓

The item is handed over to the claimant.

↓

Return Status becomes Returned.

↓

Automated email notification is sent.

↓

Process completed.

21. IMPLEMENTATION AND TESTING

The application was tested using actual records created within the ServiceNow Personal Developer Instance.

One complete test scenario involved a Samsung Galaxy S24.

Lost Item

Item Name: Samsung Galaxy S24

Category: Electronics

Color: Black

Brand: Samsung

Lost Location: GITAM Cafeteria

Record Number: LST0001010

Status: Potential Match

Found Item

Item Name: Samsung Galaxy S24

Category: Electronics

Color: Black

Brand: Samsung

Found Location: GITAM Cafeteria

Storage Location: Security Office

Brand: Samsung

Record Number: FND0001009

Claim

Claim Number: CLM0001004

The claim progressed through:

Submitted

↓

Under Verification

↓

Approved

Return

Return Number: RET0001007

Return Status was eventually updated to:

Returned

Return Date:

04 October 2026

Notification

After the return was marked as Returned, the notification flow executed successfully.

The generated email had the subject:

Your Lost & Found Item Has Been Returned

This confirmed that the complete workflow from claim approval through item return and notification was functioning correctly.

22. TEST CASES

The following test cases were performed.

Test Case	Expected Result	Status
Create Lost Item	Lost Item record created	PASS
Create Found Item	Found Item record created	PASS
Create Claim	Claim record created	PASS
Process New Claim	Status changes to Under Verification	PASS
Find Potential Match	Matching Lost Item identified	PASS
Approve Claim	Claim status becomes Approved	PASS
Create Return	Return record created	PASS
Process Return	Return becomes Ready for Pickup	PASS
Complete Return	Return status becomes Returned	PASS
Send Return Notification Email	generated	PASS
Report Lost Item Form	Record created successfully	PASS
Report Found Item Form	Record created successfully	PASS

23. RESULTS

The developed application successfully demonstrated the following capabilities:

- Centralized management of lost and found items.
- Structured claim management.
- Automated claim status processing.
- Potential matching of lost and found items.
- Automated creation of return records.
- Automated return status management.
- Automated claim approval notifications.
- Automated item-return notifications.
- Role-based access control.
- Custom portal-based reporting forms.
- End-to-end workflow automation.

The final workflow successfully connected the four major application modules and demonstrated how ServiceNow can be used to automate a real-world business process.

24. CHALLENGES FACED DURING DEVELOPMENT

24.1 Dynamic Data Mapping

One of the challenges encountered during the development of the matching flow was configuring dynamic data pills correctly.

The matching flow required the Category, Color, and Brand values from the Found Item record to be individually mapped to the corresponding fields in the Lost Items lookup.

The issue was resolved by selecting the individual nested fields rather than passing the complete Found Item record.

24.2 Workflow Testing

Each flow required individual testing before performing an end-to-end test.

The workflows were tested for:

- Correct trigger conditions.
- Correct field mapping.
- Correct status changes.
- Correct record creation.
- Correct email generation.

24.3 Portal Form Configuration

The custom reporting pages required configuration of:

- Form Controller
- Table
- Sys ID
- Form Fields
- Sections
- Save Action

Separate configurations were required for the Lost Items and Found Items tables.

25. ADVANTAGES

The Smart Lost & Found Management System provides the following advantages:

Centralized Management

All lost-and-found information is maintained within a single ServiceNow application.

Automation

Repeated administrative tasks are automated using Flow Designer.

Improved Tracking

Every record has a unique identifier, making records easier to track.

Structured Claim Processing

Claims follow a defined verification and approval process.

Potential Matching

The system can identify potential matches using item attributes.

Automated Communication

Users receive automated email notifications when important actions occur.

Role-Based Access

Different application roles provide controlled access to system functionality.

User-Friendly Interface

Custom portal forms provide a simple interface for users.

End-to-End Visibility

The complete lifecycle of an item can be tracked from reporting to return.

26. LIMITATIONS

The current implementation has the following limitations:

1. The matching mechanism currently uses basic attributes such as category, color, and brand.
2. Image-based matching has not been implemented.
3. Advanced AI-based matching has not been implemented.
4. The application is currently developed and tested in a ServiceNow Personal Developer Instance.
5. Advanced analytics and dashboards are not included in the current version.
6. External SMS or mobile push notifications are not currently implemented.

27. FUTURE ENHANCEMENTS

The system can be further enhanced using AI, analytics, and additional integrations.

AI-Based Image Matching

Computer vision can be integrated to compare images of lost and found items.

The system could generate a similarity score and identify highly similar items.

Intelligent Matching

Machine learning can be used to calculate matching scores based on:

- Category
- Color
- Brand
- Location
- Date
- Description
- Image similarity

Analytics Dashboard

A dashboard could display:

- Total Lost Items
- Total Found Items
- Potential Matches
- Pending Claims
- Approved Claims
- Returned Items
- Average Resolution Time

Mobile Application

A mobile application could allow users to report lost and found items directly from their smartphones.

QR Code Integration

QR codes could be used to simplify item identification and return processing.

Advanced Notifications

The system could be extended with SMS, push notifications, or integration with collaboration platforms.

28. SECURITY

Security was implemented through role-based access control.

The administrator role provides management capabilities, while the user role provides access to user-oriented functionality.

The system separates administrative operations from normal user interactions.

Future versions can further enhance security using:

- More granular access controls.
- Detailed audit logging.
- Stronger authentication.
- Additional data validation.

29. BUSINESS VALUE

The Smart Lost & Found Management System provides a structured approach to managing lost property.

Instead of depending on manual registers, spreadsheets, or informal communication, the organization can maintain a centralized record of every lost item, found item, claim, and return.

The automation provided by ServiceNow can help reduce repetitive administrative activities while improving process visibility and communication.

The system therefore demonstrates how an enterprise platform such as ServiceNow can be used to convert a manual operational process into a structured digital workflow.

30. CONCLUSION

The Smart Lost & Found Management System successfully demonstrates the development of a workflow-driven enterprise application using ServiceNow.

The application provides dedicated modules for Lost Items, Found Items, Claims, and Item Returns. These modules are connected through reference relationships and automated workflows.

ServiceNow Flow Designer was used to automate major business processes, including claim processing, potential item matching, return creation, return processing, and email notifications.

A custom Service Portal experience was also developed to provide users with dedicated forms for reporting lost and found items.

The complete workflow was tested using actual records in the ServiceNow Personal Developer Instance.

The final system demonstrates the practical use of:

ServiceNow App Engine Studio + Custom Tables + Reference Fields + Roles + Flow Designer + UI Builder + Service Portal + Automated Notifications

The project successfully transforms the traditional lost-and-found process into a centralized and automated digital workflow.